

Incident Handler's Journal

Completed cybersecurity portfolio document

Purpose

This journal documents incident investigation practice, cybersecurity tool usage, and reflections from the course. It is organized as a portfolio-ready record that demonstrates how security events can be captured using consistent documentation, the 5 W's, and the NIST Incident Response Lifecycle phases.

Minimum Activity Requirements Addressed

- Four dated and numbered journal entries are completed.
- Two entries document incident investigations using the 5 W's.
- Two entries describe the use of cybersecurity tools.
- The Reflections/Notes section answers the required learning prompts.

Journal Entry 1: Ransomware incident investigation - health care clinic

Description	Documented a ransomware incident affecting a small U.S. health care clinic. This entry focuses on the Detection and Analysis phase because the available facts identify the attack vector, affected assets, business impact, and initial evidence needed to support response planning.
Tool(s) used	Incident handler's journal, employee reports, endpoint alerts, system availability checks, and incident documentation notes.
The 5 W's	Who: An organized group of unethical hackers caused the incident by sending targeted phishing emails to employees. What: A malicious attachment installed malware after being downloaded. The attackers deployed ransomware that encrypted critical files and displayed a ransom note demanding payment for a decryption key. When: The incident was discovered on a Tuesday morning at approximately 9:00 a.m. Where: The incident affected the clinic's internal computer systems, including systems used to access medical records and other critical patient data. Why: The incident occurred because users received phishing emails with malicious attachments, allowing the attackers to gain initial access and deploy ransomware across affected systems.
Additional notes	The clinic should preserve evidence, isolate affected machines, notify appropriate internal stakeholders, and avoid making recovery decisions until the scope of encryption and available backups are confirmed. This incident shows how phishing can quickly become a business continuity issue when critical data becomes unavailable.

Journal Entry 2: Suspicious file hash investigation

Description	Investigated a suspicious file by reviewing its cryptographic hash and comparing it with known indicators. This entry focuses on Detection and Analysis because the goal was to determine whether the file was likely malicious before deciding what containment steps should follow.
Tool(s) used	SHA-256 hash value, VirusTotal or similar threat intelligence lookup, endpoint file metadata, and incident notes.
The 5 W's	Who: The exact actor was unknown, but the suspicious file may have been introduced by an external threat actor or through an employee downloading unsafe content. What: A suspicious file was identified and analyzed using its hash. The hash helped determine whether the file matched known malware or other previously reported suspicious files. When: The investigation occurred after the file was flagged during security review or suspicious activity triage. Where: The file was located on an endpoint or internal system being reviewed by the security team. Why: The investigation was performed because unknown files can create risk if they contain malware, persistence mechanisms, or code designed to compromise systems or data.
Additional notes	Hash-based analysis is useful because it allows analysts to check a file without opening or executing it. If the file is confirmed malicious, the next steps would include isolating the host, removing the file, checking for related indicators, and documenting lessons learned.

Journal Entry 3: Packet capture and network traffic analysis

Description	Reviewed captured network traffic to practice identifying useful packet details such as source and destination addresses, protocols, ports, timestamps, and packet behavior. This entry supports the Preparation and Detection and Analysis phases because packet analysis helps analysts recognize normal and suspicious traffic patterns.
Tool(s) used	tcpdump, Wireshark, packet capture files, protocol filters, and basic network traffic metadata.
The 5 W's	Who: No confirmed attacker was identified in this tool-focused activity. What: Network packets were captured and reviewed to understand communication patterns between systems. When: The activity occurred during the packet capture and packet analysis portion of the course. Where: Traffic was reviewed from a lab network environment or packet capture file. Why: The activity was completed to build skill in analyzing network evidence and recognizing traffic that may require deeper investigation.
Additional notes	Packet analysis is valuable because it gives direct visibility into network communication. Filtering by protocol, IP address, port, or timestamp can help reduce noise and focus an investigation on the most relevant traffic.

Journal Entry 4: SIEM and IDS log review

Description	Practiced using security monitoring tools to review alerts, logs, and event patterns. This entry supports the Detection and Analysis phase because IDS and SIEM tools help analysts identify suspicious activity, correlate events, and prioritize incidents for investigation.
Tool(s) used	Suricata alerts, Splunk queries, Chronicle searches, log fields, timestamps, source IP addresses, destination IP addresses, and event names.
The 5 W's	Who: No single actor was confirmed, but the activity involved reviewing alerts and logs that could indicate suspicious or unauthorized behavior. What: Security logs and IDS alerts were queried to identify patterns, suspicious indicators, and possible security events. When: The review occurred during course activities focused on signatures, logs, Splunk, and Chronicle. Where: The activity took place in a controlled lab environment using security monitoring data. Why: The review was performed to understand how analysts use monitoring tools to detect threats more efficiently and support incident response decisions.
Additional notes	SIEM and IDS tools are most effective when alerts are reviewed with context. Useful context includes asset criticality, baseline activity, related events, and whether the alert aligns with known attack behavior.

Reflections/Notes

Were there any specific activities that were challenging for you? Why or why not?

The most challenging activities were the ones that required connecting technical evidence to a clear incident story. It was not enough to identify a suspicious file, alert, or packet; I also had to explain what the evidence meant, why it mattered, and what response steps would be appropriate.

Has your understanding of incident detection and response changed since taking this course?

Yes. I now understand incident response as a structured process rather than a single technical action. Detection, containment, analysis, recovery, and post-incident improvement all depend on accurate documentation, evidence handling, and communication with the right stakeholders.

Was there a specific tool or concept that you enjoyed the most? Why?

I enjoyed working with SIEM and log analysis concepts the most because they connect many cybersecurity skills together. Reviewing logs, searching for indicators, and correlating events helped me see how analysts turn large amounts of activity into useful security findings.

Portfolio Summary

This completed incident handler's journal demonstrates the ability to document incidents, organize evidence, apply the 5 W's, and reflect on the use of cybersecurity tools. The entries show how structured documentation supports clearer communication, stronger investigation habits, and better preparation for real incident response work.

Completion Checklist

- Entry dates and numbers included.
- Descriptions completed in a professional tone.
- Incident investigation entries include the 5 W's.
- Cybersecurity tools are identified where applicable.
- Reflection prompts are answered clearly.